

Visual studio guidelines

In Visual Studio, you can import this folder, and before running it, there are some caveats. Please read and implement in order!

1 wget, conda, pip, brew

```
1 $ wget https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-x86_64.sh -O
  → ~/miniconda.sh
2 bash ~/miniconda.sh -b -p $HOME/miniconda
```

Other things can be found online. Feel free to add your line of terminal prompt! We proceed by assuming these are installed.

2 PyTorch

Python version and install Pytorch PyTorch is supported by Python versions below 3.13. I used 3.11. If your Python version is different, you can run the following (on Mac):

```
1 $ conda create -n clean_torch python=3.11 -y
2 $ conda activate clean_torch
3 $ python -m pip install --upgrade pip
4 $ python -m pip install torch torchvision torchaudio --index-url
  → https://download.pytorch.org/whl/cpu
```

or

```
1 $ conda create -n bcat python=3.11 -y
2 $ conda activate bcat
3 $ pip install torch torchvision torchaudio
```

PyTorch version Pytorch version: should be 2.x.x. Check if the PyTorch version is correct.

```
1 $ python -c "import torch; print(torch.__version__)"
```

3 Other libraries

Install other libraries:

```
1 $ pip install h5py
2 $ pip install numpy==1.26.3
3 $ pip install matplotlib
4 $ python -m pip install einops
5 $ python -m pip install pandas
6 $ python3 -c "import pandas as pd; print(pd.__version__)"
```

Note that Numpy should be downgraded to a version below 2!

4 Python Interpreter in VS

Now, in Visual Studio, set the correct Python version and compiler.

1. Choose the right Python version on the left panel (See the first two figures below).

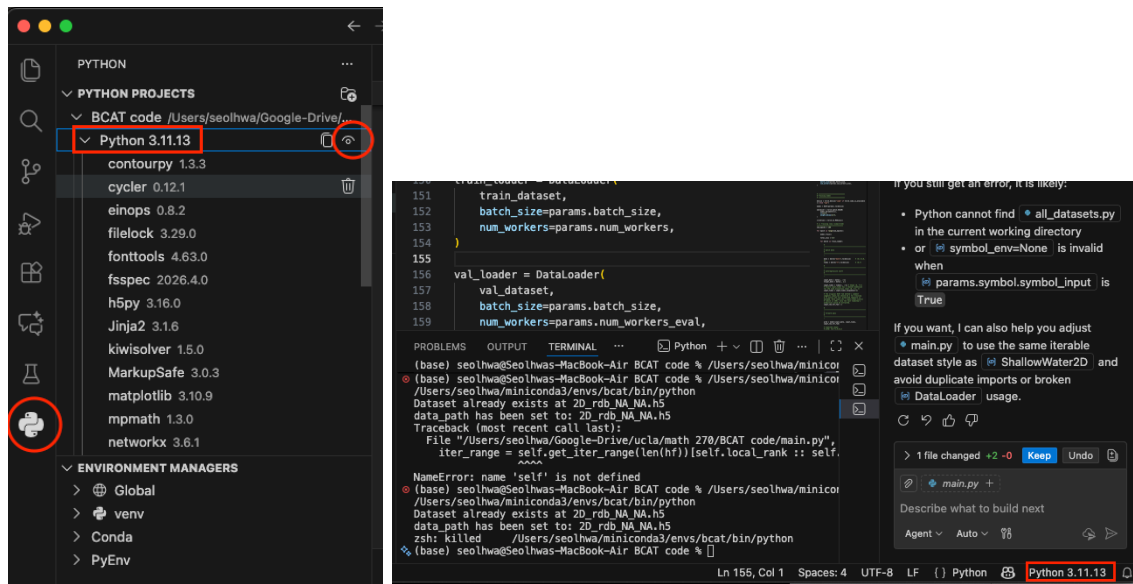


Figure 1: Enter Caption

2. Choose the correct path to the Python interpreter: In Visual Studio, **Shift+cmd+P** will display a new window in the third figure.

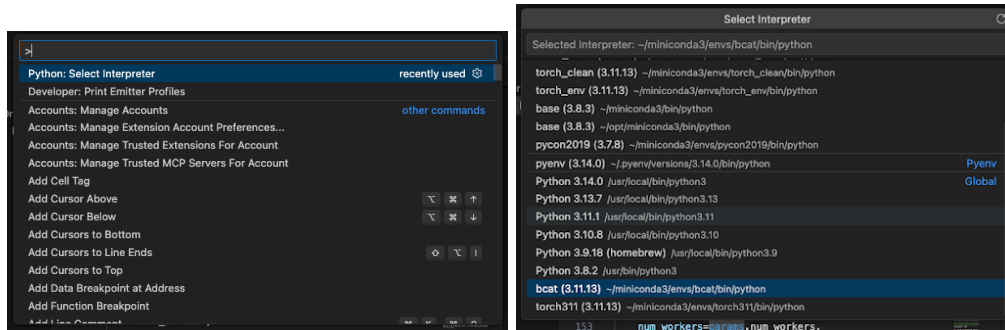


Figure 2: Enter Caption

Click Select Interpreter (the blue line in the third figure). Choose the path that has (...)/env/bcat/bin/python (blue line in last figure).

3. Run **main_test.py** for the test. If it runs, run **main.py**.

5 Cleaning Caches

The visual studio (and other applications's) caches are located at

```

1 $ ls -d "./Library/Application\ Support/Code/"* | grep -i cache
2 ./Library/Application\ Support/Code/Cache
3 ./Library/Application\ Support/Code/CachedData
4 ./Library/Application\ Support/Code/CachedExtensionVSIxS
5 ./Library/Application\ Support/Code/CachedExtensions
6 ./Library/Application\ Support/Code/Code Cache
7 ./Library/Application\ Support/Code/GPUCache

```

One can remove them using

```

1 $ rm -rf ./Library/Application\ Support/Code/Cache

```

and so on.